

Juergen Gall

Temporal Filtering Computer Vision

MA-INF 2201 / MA-MOROB-M04
WS25/26

Literature



Chapter 19 Temporal Models

S. Prince. **Computer Vision: Models, Learning, and Inference**. Cambridge University Press 2012

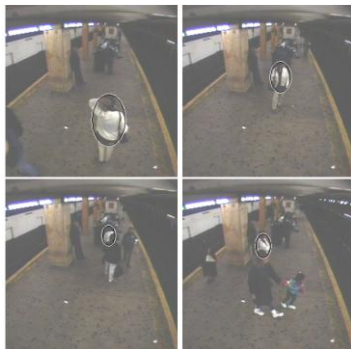
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Mean Shift Tracking

Goal: Mark object in first frame and locate it in all frames



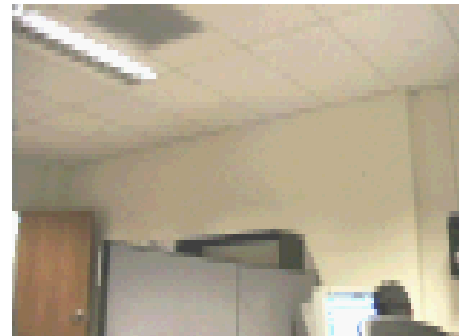
D. Comaniciu. **Real-Time Tracking of Non-Rigid Objects using Mean Shift**. CVPR 2000

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Mean shift tracking



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Goal



To track object state from frame to frame in a video

Difficulties:

- Clutter (data association)
- One image may not be enough to fully define state
- Relationship between frames may be complicated

Source: S. Prince

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Structure

- Temporal models
- Kalman filter
- Extended Kalman filter
- Unscented Kalman filter
- Particle filters
- Applications

Source: S. Prince

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Temporal Models



- Consider an evolving system
- Represented by an unknown vector, $\mathbf{w}$
- This is termed the **state**
- Examples:
 - 2D Position of tracked object in image
 - 3D Pose of tracked object in world
 - Joint positions of articulated model

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Source: S. Prince

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Estimating State



Two contributions to estimating the state:

1. A set of **measurements** $\mathbf{x}_t$, which provide information about the state $\mathbf{w}_t$ at time t . This is a generative model: the measurements are derived from the state using a known probability relation $\Pr(\mathbf{x}_t|\mathbf{w}_1...\mathbf{w}_T)$
2. A **time series model**, which says something about the expected way that the system will evolve e.g., $\Pr(\mathbf{w}_t|\mathbf{w}_1...\mathbf{w}_{t-1}, \mathbf{w}_{t+1}...\mathbf{w}_T)$

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Source: S. Prince

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Assumptions



- Only the immediate past matters (Markov)
 - the probability of the state at time t is conditionally independent of states at times $1...t-2$ given the state at time $t-1$, i.e., $\Pr(\mathbf{w}_t|\mathbf{w}_{t-1}, \dots, \mathbf{w}_1) = \Pr(\mathbf{w}_t|\mathbf{w}_{t-1})$
- Measurements depend on only the current state
 - the likelihood of the measurements at time t is conditionally independent of all of the other measurements and the states at times $1...t-1$ given the state at time t , $\Pr(\mathbf{x}_t|\mathbf{w}_t, \dots, \mathbf{w}_1, \mathbf{x}_{t-1}, \dots, \mathbf{x}_1) = \Pr(\mathbf{x}_t|\mathbf{w}_t)$

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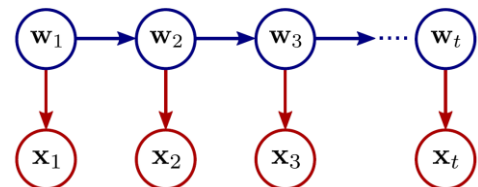
Source: S. Prince

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Graphical Model



World states



Measurements

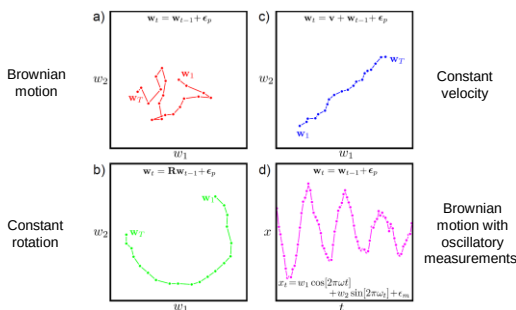
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Source: S. Prince

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Temporal Models



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Recursive Estimation



Time 1

$$\begin{aligned} \Pr(w_1|x_1) &= \frac{\Pr(x_1|w_1)\Pr(w_1)}{\Pr(x_1)} \\ &= \frac{\Pr(x_1|w_1)\Pr(w_1)}{\int \Pr(x_1|w_1)\Pr(w_1)dw_1} \end{aligned}$$

Time 2

$$\begin{aligned} \Pr(w_2|x_2, x_1) &= \frac{\Pr(x_2|w_2, x_1)\Pr(w_2|x_1)}{\Pr(x_2|x_1)} \\ &= \frac{\Pr(x_2|w_2)\Pr(w_2|x_1)}{\int \Pr(x_2|w_2, x_1)\Pr(w_2|x_1)dw_2} \\ &= \frac{\Pr(x_2|w_2)\Pr(w_2|x_1)}{\int \Pr(x_2|w_2)\Pr(w_2|x_1)dw_2} \end{aligned}$$

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Source: S. Prince

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Recursive Estimation



Time t

$$\begin{aligned}
 Pr(w_t | x_t, x_{1...t-1}) &= \frac{Pr(x_t | w_t, x_{1...t-1}) Pr(w_t | x_{1...t-1})}{Pr(x_t | x_{1...t-1})} \\
 &= \frac{Pr(x_t | w_t) Pr(w_t | x_{1...t-1})}{\int Pr(x_t | w_t) Pr(w_t | x_{1...t-1}) dw_t} \\
 &= \frac{Pr(x_t | w_t) Pr(w_t | x_{1...t-1})}{\int Pr(x_t | w_t) Pr(w_t | x_{1...t-1}) dw_t}
 \end{aligned}$$

Measurement model

Prediction from temporal model

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Computing the prior (time evolution)



Each time, the prior is based on the [Chapman-Kolmogorov](#) equation

$$Pr(\mathbf{w}_t | \mathbf{x}_{1...t-1}) = \int Pr(\mathbf{w}_t | \mathbf{w}_{t-1}) Pr(\mathbf{w}_{t-1} | \mathbf{x}_{1...t-1}) d\mathbf{w}_{t-1}$$

Prior at time t

Temporal model

Posterior at time t-1

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Summary



Alternate between:

Temporal Evolution

Temporal model

$$Pr(\mathbf{w}_t | \mathbf{x}_{1...t-1}) = \int Pr(\mathbf{w}_t | \mathbf{w}_{t-1}) Pr(\mathbf{w}_{t-1} | \mathbf{x}_{1...t-1}) d\mathbf{w}_{t-1}$$

Measurement Update

Measurement model

$$Pr(\mathbf{w}_t | \mathbf{x}_{1...t}) = \frac{Pr(\mathbf{x}_t | \mathbf{w}_t) Pr(\mathbf{w}_t | \mathbf{x}_{1...t-1})}{\int Pr(\mathbf{x}_t | \mathbf{w}_t) Pr(\mathbf{w}_t | \mathbf{x}_{1...t-1}) d\mathbf{w}_t}$$

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What is the problem?



$$Pr(\mathbf{w}_t | \mathbf{x}_{1...t-1}) = \int Pr(\mathbf{w}_t | \mathbf{w}_{t-1}) Pr(\mathbf{w}_{t-1} | \mathbf{x}_{1...t-1}) d\mathbf{w}_{t-1}$$

$$Pr(\mathbf{w}_t | \mathbf{x}_{1...t}) = \frac{Pr(\mathbf{x}_t | \mathbf{w}_t) Pr(\mathbf{w}_t | \mathbf{x}_{1...t-1})}{\int Pr(\mathbf{x}_t | \mathbf{w}_t) Pr(\mathbf{w}_t | \mathbf{x}_{1...t-1}) d\mathbf{w}_t}$$

- No closed form solution
- Integrals expensive to compute
- Standard tricks:
 - Assume the world to be linear and Gaussian (Kalman filter)
 - Approximate non-linearity by Taylor expansion (Extended Kalman filter)
 - Sampling (Particle filter)

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Structure



- Temporal models
- [Kalman filter](#)
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- Particle filters
- Applications

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Kalman Filter



The Kalman filter is just a special case of this type of recursive estimation procedure.

Temporal model and measurement model [carefully chosen](#) so that if the posterior at time t-1 was Gaussian then the

- prior at time t will be Gaussian
- posterior at time t will be Gaussian

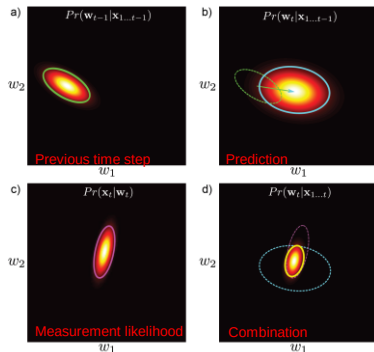
The Kalman filter equations are rules for updating the means and covariances of these Gaussians

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Source: S. Prince 17

The Kalman Filter



Source: S. Prince

Kalman Filter Definition

Time evolution equation

$$w_t = \mu_p + \Psi w_{t-1} + \epsilon_p$$

State transition matrix

Additive Gaussian noise

Measurement equation

$$x_t = \mu_m + \Phi w_t + \epsilon_m$$

Relates state and measurement

Additive Gaussian noise

Source: S. Prince

Kalman Filter Definition

Time evolution equation

$$Pr(w_t|w_{t-1}) = \text{Norm}_{w_t}[\mu_p + \Psi w_{t-1}, \Sigma_p]$$

State transition matrix

Additive Gaussian noise

Measurement equation

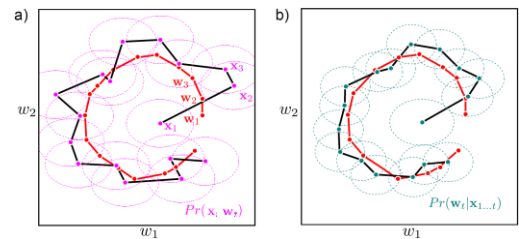
$$Pr(x_t|w_t) = \text{Norm}_{x_t}[\mu_m + \Phi w_t, \Sigma_m]$$

Relates state and measurement

Additive Gaussian noise

Source: S. Prince

Kalman Filter Example



$$Pr(w_t|w_{t-1}) = \text{Norm}_{w_t}[w_{t-1}, \sigma_p^2 \mathbf{I}] \quad \text{Not correct model!}$$

$$Pr(x_t|w_t) = \text{Norm}_{x_t}[w_t, \Sigma_m]$$

Red: True State, Magenta: Observations, Green: Estimate (Kalman)

Benefit of temporal integration: Smoother and lower covariance

Source: S. Prince

Kalman Filter Example

General:

$$Pr(w_t|w_{t-1}) = \text{Norm}_{w_t}[\mu_p + \Psi w_{t-1}, \Sigma_p]$$

$$Pr(x_t|w_t) = \text{Norm}_{x_t}[\mu_m + \Phi w_t, \Sigma_m]$$

Example:

$$Pr(w_t|w_{t-1}) = \text{Norm}_{w_t}[w_{t-1}, \sigma_p^2 \mathbf{I}]$$

$$Pr(x_t|w_t) = \text{Norm}_{x_t}[w_t, \Sigma_m]$$

$$w_t = \begin{pmatrix} w_{1,t} \\ w_{2,t} \end{pmatrix} \quad x_t = \begin{pmatrix} x_{1,t} \\ x_{2,t} \end{pmatrix}$$

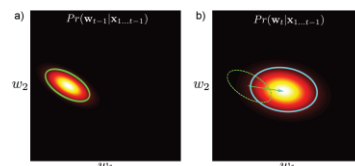
$$\mu_p = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \Psi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \Sigma_p = \begin{pmatrix} \sigma_p^2 & 0 \\ 0 & \sigma_p^2 \end{pmatrix}$$

$$\mu_m = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \Phi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \Sigma_m = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

Source: S. Prince

Temporal evolution

$$\begin{aligned} Pr(w_t|x_{1:t-1}) &= \int Pr(w_t|w_{t-1}) Pr(w_{t-1}|x_{1:t-1}) dw_{t-1} \\ &= \int \text{Norm}_{w_t}[\mu_p + \Psi w_{t-1}, \Sigma_p] \text{Norm}_{w_{t-1}}[\mu_{t-1}, \Sigma_{t-1}] dw_{t-1} \\ &= \text{Norm}_{w_t}[\mu_p + \Psi \mu_{t-1}, \Sigma_p + \Psi \Sigma_{t-1} \Psi^T] \\ &= \text{Norm}_{w_t}[\mu_+, \Sigma_+], \end{aligned}$$



Source: S. Prince

Temporal evolution

$$\begin{aligned}
 Pr(w_t | x_{1..t-1}) &= \int Pr(w_t | w_{t-1}) Pr(w_{t-1} | x_{1..t-1}) dw_{t-1} \\
 &= \int \text{Norm}_{w_t}[\mu_p + \Psi w_{t-1}, \Sigma_p] \text{Norm}_{w_{t-1}}[\mu_{t-1}, \Sigma_{t-1}] dw_{t-1} \\
 &= \int \text{Norm}_{w_t}[\mu_p + \Psi(\Psi^{-1}(w - \mu_p)), \Sigma_p] \text{Norm}_w[\Psi \mu_{t-1} + \mu_p, \Psi \Sigma_{t-1} \Psi^T] dw
 \end{aligned}$$

$w = \Psi w_{t-1} - \mu_p$
 $w_{t-1} = \Psi^{-1}(w - \mu_p)$

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Temporal evolution

$$\begin{aligned}
 Pr(w_t | x_{1..t-1}) &= \int Pr(w_t | w_{t-1}) Pr(w_{t-1} | x_{1..t-1}) dw_{t-1} \\
 &= \int \text{Norm}_{w_t}[\mu_p + \Psi w_{t-1}, \Sigma_p] \text{Norm}_{w_{t-1}}[\mu_{t-1}, \Sigma_{t-1}] dw_{t-1} \\
 &= \int \text{Norm}_{w_t}[\mu_p + \Psi(\Psi^{-1}(w - \mu_p)), \Sigma_p] \text{Norm}_w[\Psi \mu_{t-1} + \mu_p, \Psi \Sigma_{t-1} \Psi^T] dw \\
 &= \int \text{Norm}_w[w_t, \Sigma_p] \text{Norm}_w[\Psi \mu_{t-1} + \mu_p, \Psi \Sigma_{t-1} \Psi^T] dw \\
 &= \text{Norm}_{w_t}[\Psi \mu_{t-1} + \mu_p, \Sigma_p + \Psi \Sigma_{t-1} \Psi^T] \quad \boxed{\Sigma_* = (\Sigma_p^{-1} + (\Psi \Sigma_{t-1} \Psi^T)^{-1})^{-1}} \\
 &\quad \cdot \int \text{Norm}_w[\Sigma_* (\Sigma_p^{-1} w_t + (\Psi \Sigma_{t-1} \Psi^T)^{-1} (\Psi \mu_{t-1} + \mu_p)), \Sigma_*] dw \\
 &= \text{Norm}_{w_t}[\Psi \mu_{t-1} + \mu_p, \Sigma_p + \Psi \Sigma_{t-1} \Psi^T] \quad \boxed{\int \text{Norm}_x[a, A] \text{Norm}_x[b, B] dx = \text{Norm}_x[b, A+B]} \\
 &= \text{Norm}_{w_t}[\mu_+, \Sigma_+] \quad \boxed{\int \text{Norm}_x[\Sigma_* (A^{-1}a + B^{-1}b), \Sigma_*] dx, \quad \Sigma_* = (A^{-1} + B^{-1})^{-1}}
 \end{aligned}$$

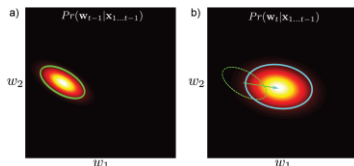
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Temporal evolution

$$\begin{aligned}
 Pr(w_t | x_{1..t-1}) &= \int Pr(w_t | w_{t-1}) Pr(w_{t-1} | x_{1..t-1}) dw_{t-1} \\
 &= \int \text{Norm}_{w_t}[\mu_p + \Psi w_{t-1}, \Sigma_p] \text{Norm}_{w_{t-1}}[\mu_{t-1}, \Sigma_{t-1}] dw_{t-1} \\
 &= \text{Norm}_{w_t}[\mu_p + \Psi \mu_{t-1}, \Sigma_p + \Psi \Sigma_{t-1} \Psi^T] \\
 &= \text{Norm}_{w_t}[\mu_+, \Sigma_+]
 \end{aligned}$$



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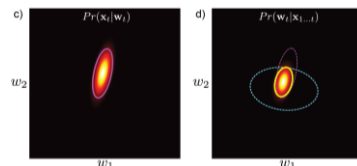
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Source: S. Prince

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Measurement incorporation

$$\begin{aligned}
 Pr(w_t | x_{1..t}) &= \frac{Pr(x_t | w_t) Pr(w_t | x_{1..t-1})}{\int Pr(x_t | w_t) Pr(w_t | x_{1..t-1}) dw_t} \\
 &= \frac{\text{Norm}_{x_t}[\mu_m + \Phi w_t, \Sigma_m] \text{Norm}_{w_t}[\mu_+, \Sigma_+]}{\int Pr(x_t | w_t) Pr(w_t | x_{1..t-1}) dw_t} \\
 &= \text{Norm}_{w_t} \left[\left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \left(\Phi^T \Sigma_m^{-1} (x_t - \mu_m) + \Sigma_+^{-1} \mu_+ \right), \right. \\
 &\quad \left. \left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \right] \\
 &= \text{Norm}_{w_t}[\mu_t, \Sigma_t]
 \end{aligned}$$



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Source: S. Prince

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Kalman Filter

$$Pr(w_t | x_{1..t}) = \text{Norm}_{w_t} \left[\left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \left(\Phi^T \Sigma_m^{-1} (x_t - \mu_m) + \Sigma_+^{-1} \mu_+ \right), \right. \\
 \left. \left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \right]$$

This is not the usual way these equations are presented.

Part of the reason for this is the size of the inverses: Φ is usually landscape and so $\Phi^T \Phi$ is inefficient.

Example:

- Measurement x : image coordinates (2d)
- State w : position + velocity (4d)
- $\Phi^T \Phi$ is 4x4 matrix
- $\Phi \Phi^T$ is 2x2 matrix

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Kalman Filter

$$Pr(w_t | x_{1..t}) = \text{Norm}_{w_t} \left[\left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \left(\Phi^T \Sigma_m^{-1} (x_t - \mu_m) + \Sigma_+^{-1} \mu_+ \right), \right. \\
 \left. \left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \right]$$

Define **Kalman gain**:

$$K = \Sigma_+ \Phi^T (\Sigma_m + \Phi \Sigma_+ \Phi^T)^{-1}$$

$$Pr(w_t | x_{1..t}) = \text{Norm}_{w_t} [\mu_+ + K(x_t - \mu_m - \Phi \mu_+), (I - K \Phi) \Sigma_+]$$

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Matrix inversion relations

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Consider the $d \times d$ matrix $\mathbf{A}$, the $k \times k$ matrix $\mathbf{C}$ and the $k \times d$ matrix $\mathbf{B}$ where $\mathbf{A}$ and $\mathbf{C}$ are symmetric, positive definite matrices. The following equality holds:

$$(\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{C}^{-1} = \mathbf{A} \mathbf{B}^T (\mathbf{B} \mathbf{A} \mathbf{B}^T + \mathbf{C})^{-1}.$$

Proof:

$$\begin{aligned} \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B} \mathbf{A} \mathbf{B}^T + \mathbf{B}^T &= \mathbf{B}^T + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B} \mathbf{A} \mathbf{B}^T \\ \mathbf{B}^T \mathbf{C}^{-1} (\mathbf{B} \mathbf{A} \mathbf{B}^T + \mathbf{C}) &= (\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B}) \mathbf{A} \mathbf{B}^T. \end{aligned}$$

Taking the inverse of both sides we get

$$(\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{C}^{-1} = \mathbf{A} \mathbf{B}^T (\mathbf{B} \mathbf{A} \mathbf{B}^T + \mathbf{C})^{-1},$$

as required.

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Matrix inversion relations

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Consider the $d \times d$ matrix $\mathbf{A}$, the $k \times k$ matrix $\mathbf{C}$ and the $k \times d$ matrix $\mathbf{B}$ where $\mathbf{A}$ and $\mathbf{C}$ are symmetric, positive definite matrices. The following equality holds:

$$(\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} = \mathbf{A} - \mathbf{A} \mathbf{B}^T (\mathbf{B} \mathbf{A} \mathbf{B}^T + \mathbf{C})^{-1} \mathbf{B} \mathbf{A}. \quad (\text{C.61})$$

This is sometimes known as the *matrix inversion lemma*.

Proof:

$$\begin{aligned} (\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} &= (\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} (\mathbf{I} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B} \mathbf{A} - \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B} \mathbf{A}) \\ &= (\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} ((\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B}) \mathbf{A} - \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B} \mathbf{A}) \\ &= \mathbf{A} - (\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B} \mathbf{A}. \end{aligned} \quad (\text{C.62})$$

We know:

$$(\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{C}^{-1} = \mathbf{A} \mathbf{B}^T (\mathbf{B} \mathbf{A} \mathbf{B}^T + \mathbf{C})^{-1}$$

Therefore:

$$(\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} = \mathbf{A} - \mathbf{A} \mathbf{B}^T (\mathbf{B} \mathbf{A} \mathbf{B}^T + \mathbf{C})^{-1} \mathbf{B} \mathbf{A}$$

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Mean Term

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$$Pr(\mathbf{w}_i | \mathbf{x}_{1:t}) = \text{Norm}_{\mathbf{w}_i} \left[\left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \left(\Phi^T \Sigma_m^{-1} (\mathbf{x}_t - \mu_m) + \Sigma_+^{-1} \mu_+ \right), \right. \\ \left. \left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \right]$$

Kalman gain:

$$\mathbf{K} = \Sigma_+ \Phi^T (\Sigma_m + \Phi \Sigma_+ \Phi^T)^{-1}$$

Matrix inversion relations:

$$(\Sigma_+^{-1} + \Phi^T \Sigma_m^{-1} \Phi)^{-1} \Phi^T \Sigma_m^{-1} = \Sigma_+ \Phi^T (\Phi \Sigma_+ \Phi^T + \Sigma_m)^{-1} = \mathbf{K}$$

$$(\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} \mathbf{B}^T \mathbf{C}^{-1} = \mathbf{A} \mathbf{B}^T (\mathbf{B} \mathbf{A} \mathbf{B}^T + \mathbf{C})^{-1}$$

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Mean Term

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$$Pr(\mathbf{w}_i | \mathbf{x}_{1:t}) = \text{Norm}_{\mathbf{w}_i} \left[\left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \left(\Phi^T \Sigma_m^{-1} (\mathbf{x}_t - \mu_m) + \Sigma_+^{-1} \mu_+ \right), \right. \\ \left. \left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \right]$$

Kalman gain:

$$\mathbf{K} = \Sigma_+ \Phi^T (\Sigma_m + \Phi \Sigma_+ \Phi^T)^{-1}$$

Matrix inversion relations:

$$(\Sigma_+^{-1} + \Phi^T \Sigma_m^{-1} \Phi)^{-1} \Phi^T \Sigma_m^{-1} = \Sigma_+ \Phi^T (\Phi \Sigma_+ \Phi^T + \Sigma_m)^{-1} = \mathbf{K}$$

$$(\Sigma_+^{-1} + \Phi^T \Sigma_m^{-1} \Phi)^{-1} = \Sigma_+ - \Sigma_+ \Phi^T (\Phi \Sigma_+ \Phi^T + \Sigma_m)^{-1} \Phi \Sigma_+$$

$$(\mathbf{A}^{-1} + \mathbf{B}^T \mathbf{C}^{-1} \mathbf{B})^{-1} = \mathbf{A} - \mathbf{A} \mathbf{B}^T (\mathbf{B} \mathbf{A} \mathbf{B}^T + \mathbf{C})^{-1} \mathbf{B} \mathbf{A}.$$

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Mean Term

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$$Pr(\mathbf{w}_i | \mathbf{x}_{1:t}) = \text{Norm}_{\mathbf{w}_i} \left[\left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \left(\Phi^T \Sigma_m^{-1} (\mathbf{x}_t - \mu_m) + \Sigma_+^{-1} \mu_+ \right), \right. \\ \left. \left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \right]$$

Kalman gain:

$$\mathbf{K} = \Sigma_+ \Phi^T (\Sigma_m + \Phi \Sigma_+ \Phi^T)^{-1}$$

Matrix inversion relations:

$$(\Sigma_+^{-1} + \Phi^T \Sigma_m^{-1} \Phi)^{-1} \Phi^T \Sigma_m^{-1} = \Sigma_+ \Phi^T (\Phi \Sigma_+ \Phi^T + \Sigma_m)^{-1} = \mathbf{K}$$

$$(\Sigma_+^{-1} + \Phi^T \Sigma_m^{-1} \Phi)^{-1} = \Sigma_+ - \Sigma_+ \Phi^T (\Phi \Sigma_+ \Phi^T + \Sigma_m)^{-1} \Phi \Sigma_+$$

Using Matrix inversion relations:

$$\begin{aligned} (\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1})^{-1} (\Phi^T \Sigma_m^{-1} (\mathbf{x}_t - \mu_m) + \Sigma_+^{-1} \mu_+) &= \mathbf{K} (\mathbf{x}_t - \mu_m) + (\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1})^{-1} \Sigma_+^{-1} \mu_+ \\ &= \mathbf{K} (\mathbf{x}_t - \mu_m) + (\Sigma_+ - \Sigma_+ \Phi^T (\Phi \Sigma_+ \Phi^T + \Sigma_m)^{-1} \Phi \Sigma_+) \Sigma_+^{-1} \mu_+ \\ &= \mathbf{K} (\mathbf{x}_t - \mu_m) + \mu_+ - \Sigma_+ \Phi^T (\Phi \Sigma_+ \Phi^T + \Sigma_m)^{-1} \Phi \Sigma_+ \mu_+ \\ &= \mathbf{K} (\mathbf{x}_t - \mu_m) + \mu_+ - \mathbf{K} \Phi \mu_+ \\ &= \mu_+ + \mathbf{K} (\mathbf{x}_t - \mu_m - \Phi \mu_+) \end{aligned}$$

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Covariance Term

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$$Pr(\mathbf{w}_i | \mathbf{x}_{1:t}) = \text{Norm}_{\mathbf{w}_i} \left[\left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \left(\Phi^T \Sigma_m^{-1} (\mathbf{x}_t - \mu_m) + \Sigma_+^{-1} \mu_+ \right), \right. \\ \left. \left(\Phi^T \Sigma_m^{-1} \Phi + \Sigma_+^{-1} \right)^{-1} \right]$$

Kalman gain:

$$\mathbf{K} = \Sigma_+ \Phi^T (\Sigma_m + \Phi \Sigma_+ \Phi^T)^{-1}$$

Matrix inversion relations:

$$(\Sigma_+^{-1} + \Phi^T \Sigma_m^{-1} \Phi)^{-1} = \Sigma_+ - \Sigma_+ \Phi^T (\Phi \Sigma_+ \Phi^T + \Sigma_m)^{-1} \Phi \Sigma_+$$

$$= \Sigma_+ - \mathbf{K} \Phi \Sigma_+$$

$$= (\mathbf{I} - \mathbf{K} \Phi) \Sigma_+$$

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Final Kalman Filter Equation

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$$Pr(\mathbf{w}_t | \mathbf{x}_{1:t}) = \text{Norm}_{\mathbf{w}_t} [\boldsymbol{\mu}_+ + \mathbf{K}(\mathbf{x}_t - \boldsymbol{\mu}_m - \boldsymbol{\Phi}\boldsymbol{\mu}_+), (\mathbf{I} - \mathbf{K}\boldsymbol{\Phi})\boldsymbol{\Sigma}_+]$$

Innovation (difference between actual and predicted measurements)

Prior variance minus a term due to information from measurement

18.12.2025 Source: S. Prince

Kalman Filter Summary

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Time evolution equation

$$Pr(\mathbf{w}_t | \mathbf{w}_{t-1}) = \text{Norm}_{\mathbf{w}_t} [\boldsymbol{\mu}_p + \boldsymbol{\Psi}\mathbf{w}_{t-1}, \boldsymbol{\Sigma}_p]$$

Measurement equation

$$Pr(\mathbf{x}_t | \mathbf{w}_t) = \text{Norm}_{\mathbf{x}_t} [\boldsymbol{\mu}_m + \boldsymbol{\Phi}\mathbf{w}_t, \boldsymbol{\Sigma}_m]$$

Inference

$$\text{State Prediction: } \boldsymbol{\mu}_+ = \boldsymbol{\mu}_p + \boldsymbol{\Psi}\boldsymbol{\mu}_{t-1}$$

$$\text{Covariance Prediction: } \boldsymbol{\Sigma}_+ = \boldsymbol{\Sigma}_p + \boldsymbol{\Psi}\boldsymbol{\Sigma}_{t-1}\boldsymbol{\Psi}^T$$

$$\text{State Update: } \boldsymbol{\mu}_t = \boldsymbol{\mu}_+ + \mathbf{K}(\mathbf{x}_t - \boldsymbol{\mu}_m - \boldsymbol{\Phi}\boldsymbol{\mu}_+)$$

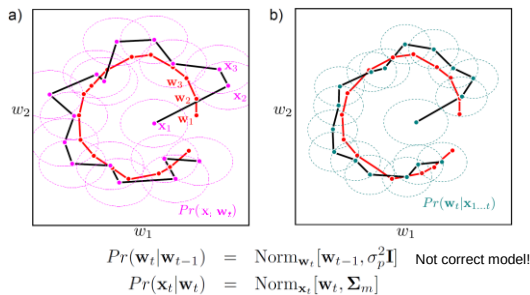
$$\text{Covariance Update: } \boldsymbol{\Sigma}_t = (\mathbf{I} - \mathbf{K}\boldsymbol{\Phi})\boldsymbol{\Sigma}_+,$$

$$\mathbf{K} = \boldsymbol{\Sigma}_+ \boldsymbol{\Phi}^T (\boldsymbol{\Sigma}_m + \boldsymbol{\Phi}\boldsymbol{\Sigma}_+ \boldsymbol{\Phi}^T)^{-1}$$

18.12.2025 Source: S. Prince

Kalman Filter Example 1

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Red: True State, Magenta: Observations, Green: Estimate (Kalman)
Benefit of temporal integration: Smoother and lower covariance

18.12.2025 Source: S. Prince

Kalman Filter Example

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General:

$$Pr(\mathbf{w}_t | \mathbf{w}_{t-1}) = \text{Norm}_{\mathbf{w}_t} [\boldsymbol{\mu}_p + \boldsymbol{\Psi}\mathbf{w}_{t-1}, \boldsymbol{\Sigma}_p]$$

$$Pr(\mathbf{x}_t | \mathbf{w}_t) = \text{Norm}_{\mathbf{x}_t} [\boldsymbol{\mu}_m + \boldsymbol{\Phi}\mathbf{w}_t, \boldsymbol{\Sigma}_m]$$

Example:

$$Pr(\mathbf{w}_t | \mathbf{w}_{t-1}) = \text{Norm}_{\mathbf{w}_t} [\mathbf{w}_{t-1}, \sigma_p^2 \mathbf{I}]$$

$$Pr(\mathbf{x}_t | \mathbf{w}_t) = \text{Norm}_{\mathbf{x}_t} [\mathbf{w}_t, \boldsymbol{\Sigma}_m]$$

$$\mathbf{w}_t = \begin{pmatrix} w_{1,t} \\ w_{2,t} \end{pmatrix} \quad \mathbf{x}_t = \begin{pmatrix} x_{1,t} \\ x_{2,t} \end{pmatrix}$$

$$\boldsymbol{\mu}_p = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \boldsymbol{\Psi} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \boldsymbol{\Sigma}_p = \begin{pmatrix} \sigma_p^2 & 0 \\ 0 & \sigma_p^2 \end{pmatrix}$$

$$\boldsymbol{\mu}_m = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \boldsymbol{\Phi} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \boldsymbol{\Sigma}_m = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

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Kalman Filter Example

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$$\text{State Prediction: } \boldsymbol{\mu}_+ = \boldsymbol{\mu}_p + \boldsymbol{\Psi}\boldsymbol{\mu}_{t-1}$$

$$\begin{pmatrix} \mu_{1,+} \\ \mu_{2,+} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \mu_{1,t-1} \\ \mu_{2,t-1} \end{pmatrix} = \begin{pmatrix} \mu_{1,t-1} \\ \mu_{2,t-1} \end{pmatrix}$$

$$\text{Covariance Prediction: } \boldsymbol{\Sigma}_+ = \boldsymbol{\Sigma}_p + \boldsymbol{\Psi}\boldsymbol{\Sigma}_{t-1}\boldsymbol{\Psi}^T$$

$$\begin{pmatrix} \Sigma_{11,+} & \Sigma_{12,+} \\ \Sigma_{21,+} & \Sigma_{22,+} \end{pmatrix} = \begin{pmatrix} \sigma_p^2 & 0 \\ 0 & \sigma_p^2 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \Sigma_{11,t-1} & \Sigma_{12,t-1} \\ \Sigma_{21,t-1} & \Sigma_{22,t-1} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \Sigma_{11,t-1} + \sigma_p^2 & \Sigma_{12,t-1} \\ \Sigma_{21,t-1} & \Sigma_{22,t-1} + \sigma_p^2 \end{pmatrix}$$

$$\text{Kalman Gain: } \mathbf{K} = \boldsymbol{\Sigma}_+ \boldsymbol{\Phi}^T (\boldsymbol{\Sigma}_m + \boldsymbol{\Phi}\boldsymbol{\Sigma}_+ \boldsymbol{\Phi}^T)^{-1}$$

$$\begin{pmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{pmatrix} = \begin{pmatrix} \Sigma_{11,+} & \Sigma_{12,+} \\ \Sigma_{21,+} & \Sigma_{22,+} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \left(\begin{pmatrix} \Sigma_{11,m} & \Sigma_{12,m} \\ \Sigma_{21,m} & \Sigma_{22,m} \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \Sigma_{11,+} & \Sigma_{12,+} \\ \Sigma_{21,+} & \Sigma_{22,+} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right)^{-1}$$

$$= \begin{pmatrix} \Sigma_{11,+} & \Sigma_{12,+} \\ \Sigma_{21,+} & \Sigma_{22,+} \end{pmatrix} \begin{pmatrix} \Sigma_{11,m} + \Sigma_{11,+} & \Sigma_{12,m} + \Sigma_{12,+} \\ \Sigma_{21,m} + \Sigma_{21,+} & \Sigma_{22,m} + \Sigma_{22,+} \end{pmatrix}^{-1}$$

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Kalman Filter Example

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$$\text{State Update: } \boldsymbol{\mu}_t = \boldsymbol{\mu}_+ + \mathbf{K}(\mathbf{x}_t - \boldsymbol{\mu}_m - \boldsymbol{\Phi}\boldsymbol{\mu}_+)$$

$$\begin{pmatrix} \mu_{1,t} \\ \mu_{2,t} \end{pmatrix} = \begin{pmatrix} \mu_{1,+} \\ \mu_{2,+} \end{pmatrix} + \begin{pmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{pmatrix} \left(\begin{pmatrix} x_{1,t} \\ x_{2,t} \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \mu_{1,+} \\ \mu_{2,+} \end{pmatrix} \right)$$

$$= \begin{pmatrix} \mu_{1,+} \\ \mu_{2,+} \end{pmatrix} + \begin{pmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{pmatrix} \begin{pmatrix} x_{1,t} - \mu_{1,+} \\ x_{2,t} - \mu_{2,+} \end{pmatrix}$$

$$\text{Covariance Update: } \boldsymbol{\Sigma}_t = (\mathbf{I} - \mathbf{K}\boldsymbol{\Phi})\boldsymbol{\Sigma}_+$$

$$\begin{pmatrix} \Sigma_{11,t} & \Sigma_{12,t} \\ \Sigma_{21,t} & \Sigma_{22,t} \end{pmatrix} = \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) \begin{pmatrix} \Sigma_{11,+} & \Sigma_{12,+} \\ \Sigma_{21,+} & \Sigma_{22,+} \end{pmatrix}$$

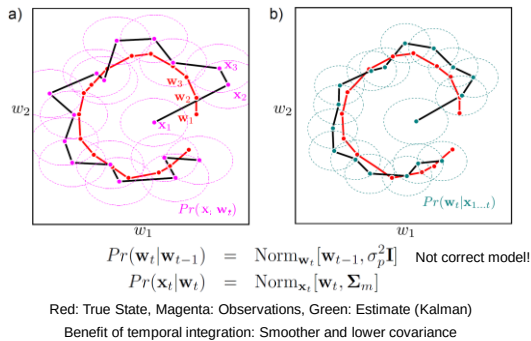
$$= \begin{pmatrix} 1 - K_{11} & -K_{12} \\ -K_{21} & 1 - K_{22} \end{pmatrix} \begin{pmatrix} \Sigma_{11,+} & \Sigma_{12,+} \\ \Sigma_{21,+} & \Sigma_{22,+} \end{pmatrix}$$

Except of one matrix inversion, very easy to compute!

18.12.2025 Source: S. Prince

Kalman Filter Example 1

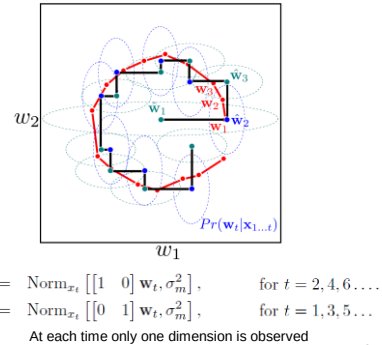
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Kalman Filter Example 2

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Smoothing

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- Estimates depend only on measurements up to the current point in time.
- Sometimes want to estimate state based on future measurements as well

Fixed Lag Smoother:

This is an on-line scheme in which the optimal estimate for a state at time $t - \tau$ is calculated based on measurements up to time t , where τ is the time lag, i.e. we wish to calculate $Pr(\mathbf{w}_{t-\tau} | \mathbf{x}_1 \dots \mathbf{x}_t)$.

Fixed Interval Smoother:

We have a fixed time interval of measurements and want to calculate the optimal state estimate based on all of these measurements. In other words, instead of calculating $Pr(\mathbf{w}_t | \mathbf{x}_1 \dots \mathbf{x}_t)$ we now estimate $Pr(\mathbf{w}_t | \mathbf{x}_1 \dots \mathbf{x}_T)$ where T is the total length of the interval.

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Fixed lag smoother

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State evolution equation

$$\begin{bmatrix} \mathbf{w}_t^{[1]} \\ \mathbf{w}_t^{[2]} \\ \vdots \\ \mathbf{w}_t^{[\tau]} \end{bmatrix} = \begin{bmatrix} \Psi & 0 & \dots & 0 & 0 \\ \mathbf{I} & 0 & \dots & 0 & 0 \\ 0 & \mathbf{I} & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & \mathbf{I} & 0 \end{bmatrix} \begin{bmatrix} \mathbf{w}_{t-1}^{[1]} \\ \mathbf{w}_{t-1}^{[2]} \\ \vdots \\ \mathbf{w}_{t-1}^{[\tau]} \end{bmatrix} + \begin{bmatrix} \epsilon_p \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Estimate delayed by τ →

Measurement equation

$$\mathbf{x}_t = \begin{bmatrix} \Phi & 0 & 0 & \dots & 0 \end{bmatrix} \begin{bmatrix} \mathbf{w}_t^{[1]} \\ \mathbf{w}_t^{[2]} \\ \vdots \\ \mathbf{w}_t^{[\tau]} \end{bmatrix} + \epsilon_m$$

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Kalman Filter Summary

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Time evolution equation

$$Pr(\mathbf{w}_t | \mathbf{w}_{t-1}) = \text{Norm}_{\mathbf{w}_t}[\mu_p + \Psi \mathbf{w}_{t-1}, \Sigma_p]$$

Measurement equation

$$Pr(\mathbf{x}_t | \mathbf{w}_t) = \text{Norm}_{\mathbf{x}_t}[\mu_m + \Phi \mathbf{w}_t, \Sigma_m]$$

Inference

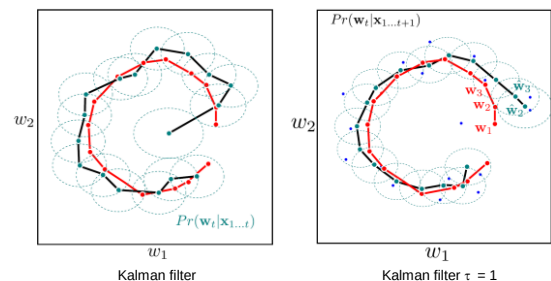
State Prediction: $\mu_+ = \mu_p + \Psi \mu_{t-1}$
 Covariance Prediction: $\Sigma_+ = \Sigma_p + \Psi \Sigma_{t-1} \Psi^T$
 State Update: $\mu_t = \mu_+ + \mathbf{K}(\mathbf{x}_t - \mu_m - \Phi \mu_+)$
 Covariance Update: $\Sigma_t = (\mathbf{I} - \mathbf{K} \Phi) \Sigma_+$

$$\mathbf{K} = \Sigma_+ \Phi^T (\Sigma_m + \Phi \Sigma_+ \Phi^T)^{-1}$$

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Fixed-lag Kalman Smoothing

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Advantage: Better estimates and lower covariance
 Disadvantage: Delay

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Fixed interval smoothing



Having μ_t , Σ_t , $\mu_{+|t}$ and $\Sigma_{+|t}$ from forward pass where $\Pr(w_{t+1}|x_{1:t}) = \text{Norm}(\mu_{+|t}, \Sigma_{+|t})$

Backward set of recursions

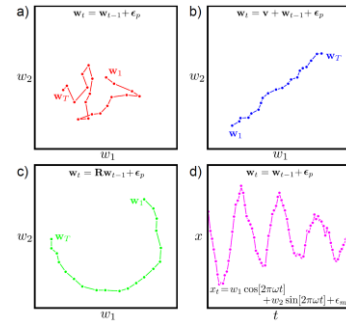
$$\begin{aligned}\mu_{t|T} &= \mu_t + C_t(\mu_{t+1|T} - \mu_{+|t}) \\ \Sigma_{t|T} &= \Sigma_t + C_t(\Sigma_{t+1|T} - \Sigma_{+|t})C_t^T\end{aligned}$$

where

$$C_t = \Sigma_{t|} \Psi^T \Sigma_{+|t}^{-1}$$

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Temporal Models



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Learning parameters



Parameters of equations:

$$\begin{aligned}w_t &= \Psi w_{t-1} + \epsilon_p \\ x_t &= \Phi w_t + \epsilon_m\end{aligned} \quad \theta = \{\Psi, \Sigma_p, \Phi, \Sigma_m, \mu_0, \Sigma_0\}$$

Expectation-Maximization

- E step:
 - Fixed interval smoothing to get (w_t)
- M step:
 - Maximize log-likelihood to get θ

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M step



Equations:

$$\begin{aligned}w_t &= \Psi w_{t-1} + \epsilon_p \\ x_t &= \Phi w_t + \epsilon_m\end{aligned}$$

Solve where μ are estimates for w (from E step):

$$\begin{aligned}\hat{\theta} &= \arg\max_{\theta} \log P(\mu, x | \theta) \quad P(\mu, x) = P(\mu_1) \prod_{t=2}^T P(\mu_t | \mu_{t-1}) \prod_{t=1}^T P(x_t | \mu_t) \\ Q &= \log P(\mu, x) = - \sum_{t=1}^T \left(\frac{1}{2} (x_t - \Phi \mu_t)^T \Sigma_m^{-1} (x_t - \Phi \mu_t) \right) - \frac{T}{2} \log |\Sigma_m| \\ &\quad - \sum_{t=2}^T \left(\frac{1}{2} (\mu_t - \Psi \mu_{t-1})^T \Sigma_p^{-1} (\mu_t - \Psi \mu_{t-1}) \right) - \frac{T-1}{2} \log |\Sigma_p| \\ &\quad - \left(\frac{1}{2} (\mu_1 - \mu_0)^T \Sigma_0^{-1} (\mu_1 - \mu_0) \right) - \frac{1}{2} \log |\Sigma_0| + \text{const.}\end{aligned}$$

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M step



$$\begin{aligned}\frac{\partial}{\partial \Phi} Q &= \frac{\partial}{\partial \Phi} \left\{ - \sum_{t=1}^T \left(\frac{1}{2} (x_t - \Phi \mu_t)^T \Sigma_m^{-1} (x_t - \Phi \mu_t) \right) \right\} \\ &= \sum_{t=1}^T \Sigma_m^{-1} (x_t - \Phi \mu_t) \mu_t^T\end{aligned}$$

$$\frac{\partial (Xb + c)^T D (Xb + c)}{\partial X} = (D + D^T)(Xb + c)b^T.$$

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M step



$$\begin{aligned}\frac{\partial}{\partial \Phi} Q &= \frac{\partial}{\partial \Phi} \left\{ - \sum_{t=1}^T \left(\frac{1}{2} (x_t - \Phi \mu_t)^T \Sigma_m^{-1} (x_t - \Phi \mu_t) \right) \right\} \\ &= \sum_{t=1}^T \Sigma_m^{-1} (x_t - \Phi \mu_t) \mu_t^T = \Sigma_m^{-1} \left(\sum_{t=1}^T x_t \mu_t^T - \Phi \sum_{t=1}^T \mu_t \mu_t^T \right) = 0 \\ \Phi &= \frac{\sum_{t=1}^T x_t \mu_t^T}{\sum_{t=1}^T \mu_t \mu_t^T}\end{aligned}$$

In the same way:

$$\Psi = \frac{\sum_{t=2}^T \mu_t \mu_{t-1}^T}{\sum_{t=2}^T \mu_{t-1} \mu_{t-1}^T}$$

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M step

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$$\begin{aligned}\frac{\partial}{\partial \Sigma_m^{-1}} Q &= \frac{\partial}{\partial \Sigma_m^{-1}} \left\{ -\sum_{t=1}^T \left(\frac{1}{2} (x_t - \Phi \mu_t)^T \Sigma_m^{-1} (x_t - \Phi \mu_t) \right) + \frac{T}{2} \log |\Sigma_m^{-1}| \right\} \\ &= -\sum_{t=1}^T \left(\frac{1}{2} (x_t - \Phi \mu_t)(x_t - \Phi \mu_t)^T \right) + \frac{T}{2} \Sigma_m \\ \frac{\partial \mathbf{a}^T \mathbf{X} \mathbf{b}}{\partial \mathbf{X}} &= \mathbf{a} \mathbf{b}^T \quad \frac{\partial}{\partial A} \log(|A|) = (A^{-1})^T\end{aligned}$$

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M step

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$$\begin{aligned}\frac{\partial}{\partial \Sigma_m^{-1}} Q &= \frac{\partial}{\partial \Sigma_m^{-1}} \left\{ -\sum_{t=1}^T \left(\frac{1}{2} (x_t - \Phi \mu_t)^T \Sigma_m^{-1} (x_t - \Phi \mu_t) \right) + \frac{T}{2} \log |\Sigma_m^{-1}| \right\} \\ &= -\sum_{t=1}^T \left(\frac{1}{2} (x_t - \Phi \mu_t)(x_t - \Phi \mu_t)^T \right) + \frac{T}{2} \Sigma_m \\ &= -\left(\sum_{t=1}^T x_t x_t^T - 2 \sum_{t=1}^T \Phi \mu_t x_t^T + \Phi \left(\sum_{t=1}^T \mu_t \mu_t^T \right) \Phi^T \right) + T \Sigma_m \\ &= -\left(\sum_{t=1}^T x_t x_t^T - 2 \sum_{t=1}^T \Phi \mu_t x_t^T + \Phi \sum_{t=1}^T \mu_t x_t^T \right) + T \Sigma_m = 0 \\ \Sigma_m &= \frac{1}{T} \sum_{t=1}^T (x_t x_t^T - \Phi \mu_t x_t^T) \\ \Sigma_p &= \frac{1}{T-1} \sum_{t=2}^T (\mu_t \mu_t^T - \mu_{t-1} \mu_{t-1}^T)\end{aligned}$$

$$\Phi = \frac{\sum_{t=1}^T x_t \mu_t^T}{\sum_{t=1}^T \mu_t \mu_t^T}$$

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Problems with the Kalman filter

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- Requires linear temporal and measurement equations
- Represents result as a normal distribution: what if the posterior is genuinely multi-modal?

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Structure

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- Temporal models
- Kalman filter
- **Extended Kalman filter**
- Unscented Kalman filter
- Particle filters
- Applications

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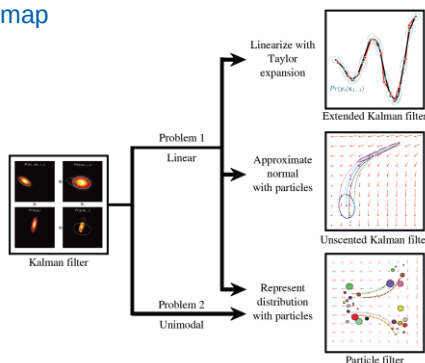
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Roadmap

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Extended Kalman Filter

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Allows non-linear measurement and temporal equations

$$\begin{aligned}\mathbf{w}_t &= \mathbf{f}[\mathbf{w}_{t-1}, \epsilon_p] \\ \mathbf{x}_t &= \mathbf{g}[\mathbf{w}_t, \epsilon_m]\end{aligned}$$

Key idea: take Taylor expansion and treat as locally linear

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Jacobians



Based on Jacobians matrices of derivatives

$$\begin{aligned}\Psi &= \left. \frac{\partial f[\mathbf{w}_{t-1}, \epsilon_p]}{\partial \mathbf{w}_{t-1}} \right|_{\mu_{t-1}, 0} \\ \Upsilon_p &= \left. \frac{\partial f[\mathbf{w}_{t-1}, \epsilon_p]}{\partial \epsilon_p} \right|_{\mu_{t-1}, 0} \\ \Phi &= \left. \frac{\partial g[\mathbf{w}_t, \epsilon_m]}{\partial \mathbf{w}_t} \right|_{\mu_+, 0} \\ \Upsilon_m &= \left. \frac{\partial g[\mathbf{w}_t, \epsilon_m]}{\partial \epsilon_m} \right|_{\mu_+, 0},\end{aligned}$$

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Extended Kalman Filter Equations



$$\begin{aligned}\text{State Prediction:} & \quad \mu_+ = f[\mu_{t-1}, 0] \\ \text{Covariance Prediction:} & \quad \Sigma_+ = \Psi \Sigma_{t-1} \Psi^T + \Upsilon_p \Sigma_p \Upsilon_p^T \\ \text{State Update:} & \quad \mu_t = \mu_+ + K(\mathbf{x}_t - g[\mu_+, 0]) \\ \text{Covariance Update:} & \quad \Sigma_t = (\mathbf{I} - K\Phi) \Sigma_+, \end{aligned}$$

where

$$K = \Sigma_+ \Phi^T (\Upsilon_m \Sigma_m \Upsilon_m^T + \Phi \Sigma_+ \Phi^T)^{-1}.$$

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Comparison: Kalman Filter



Time evolution equation

$$Pr(\mathbf{w}_t | \mathbf{w}_{t-1}) = \text{Norm}_{\mathbf{w}_t} [\mu_p + \Psi \mathbf{w}_{t-1}, \Sigma_p]$$

Measurement equation

$$Pr(\mathbf{x}_t | \mathbf{w}_t) = \text{Norm}_{\mathbf{x}_t} [\mu_m + \Phi \mathbf{w}_t, \Sigma_m]$$

Inference

$$\begin{aligned}\text{State Prediction:} & \quad \mu_+ = \mu_p + \Psi \mu_{t-1} \\ \text{Covariance Prediction:} & \quad \Sigma_+ = \Sigma_p + \Psi \Sigma_{t-1} \Psi^T \\ \text{State Update:} & \quad \mu_t = \mu_+ + K(\mathbf{x}_t - \mu_m - \Phi \mu_+) \\ \text{Covariance Update:} & \quad \Sigma_t = (\mathbf{I} - K\Phi) \Sigma_+, \end{aligned}$$

$$K = \Sigma_+ \Phi^T (\Sigma_m + \Phi \Sigma_+ \Phi^T)^{-1}$$

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Extended Kalman filter



$$\begin{aligned}\mathbf{w}_t &= f(\mathbf{w}_{t-1}, \epsilon_p) = \begin{pmatrix} w_{1,t-1} + \epsilon_{1,p} \\ w_{1,t-1} \sin(w_{1,t-1}) + \epsilon_{2,p} \end{pmatrix} \\ \mathbf{x}_t &= g(\mathbf{w}_t, \epsilon_m) = \mathbf{w}_t + \epsilon_m\end{aligned}$$

$$\begin{aligned}\Psi &= \left. \frac{\partial}{\partial \mathbf{w}} \begin{pmatrix} w_1 + \epsilon_{1,p} \\ w_1 \sin(w_1) + \epsilon_{2,p} \end{pmatrix} \right|_{\mu_{t-1}, 0} \\ \Upsilon_p &= \left. \frac{\partial}{\partial \epsilon_p} \begin{pmatrix} w_1 + \epsilon_{1,p} \\ w_1 \sin(w_1) + \epsilon_{2,p} \end{pmatrix} \right|_{\mu_{t-1}, 0} \\ \Phi &= \left. \frac{\partial}{\partial \mathbf{w}} \begin{pmatrix} w_1 + \epsilon_{1,m} \\ w_2 + \epsilon_{2,m} \end{pmatrix} \right|_{\mu_+, 0} \\ \Upsilon_m &= \left. \frac{\partial}{\partial \epsilon_m} \begin{pmatrix} w_1 + \epsilon_{1,m} \\ w_2 + \epsilon_{2,m} \end{pmatrix} \right|_{\mu_+, 0}\end{aligned}$$

$$\mathbf{y} = f(\mathbf{x}) \quad \frac{\partial f}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_M} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_N}{\partial x_1} & \cdots & \frac{\partial f_N}{\partial x_M} \end{pmatrix}$$

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Extended Kalman filter



$$\begin{aligned}\mathbf{w}_t &= f(\mathbf{w}_{t-1}, \epsilon_p) = \begin{pmatrix} w_{1,t-1} + \epsilon_{1,p} \\ w_{1,t-1} \sin(w_{1,t-1}) + \epsilon_{2,p} \end{pmatrix} \\ \mathbf{x}_t &= g(\mathbf{w}_t, \epsilon_m) = \mathbf{w}_t + \epsilon_m\end{aligned}$$

$$\begin{aligned}\Psi &= \left. \frac{\partial}{\partial \mathbf{w}} \begin{pmatrix} w_1 + \epsilon_{1,p} \\ w_1 \sin(w_1) + \epsilon_{2,p} \end{pmatrix} \right|_{\mu_{t-1}, 0} = \begin{pmatrix} 1 & 0 \\ \sin(\mu_{1,t-1}) + \mu_{1,t-1} \cos(\mu_{1,t-1}) & 0 \end{pmatrix} \\ \Upsilon_p &= \left. \frac{\partial}{\partial \epsilon_p} \begin{pmatrix} w_1 + \epsilon_{1,p} \\ w_1 \sin(w_1) + \epsilon_{2,p} \end{pmatrix} \right|_{\mu_{t-1}, 0} \\ \Phi &= \left. \frac{\partial}{\partial \mathbf{w}} \begin{pmatrix} w_1 + \epsilon_{1,m} \\ w_2 + \epsilon_{2,m} \end{pmatrix} \right|_{\mu_+, 0} \\ \Upsilon_m &= \left. \frac{\partial}{\partial \epsilon_m} \begin{pmatrix} w_1 + \epsilon_{1,m} \\ w_2 + \epsilon_{2,m} \end{pmatrix} \right|_{\mu_+, 0}\end{aligned}$$

$$\mathbf{y} = f(\mathbf{x}) \quad \frac{\partial f}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_M} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_N}{\partial x_1} & \cdots & \frac{\partial f_N}{\partial x_M} \end{pmatrix}$$

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Extended Kalman filter



$$\begin{aligned}\mathbf{w}_t &= f(\mathbf{w}_{t-1}, \epsilon_p) = \begin{pmatrix} w_{1,t-1} + \epsilon_{1,p} \\ w_{1,t-1} \sin(w_{1,t-1}) + \epsilon_{2,p} \end{pmatrix} \\ \mathbf{x}_t &= g(\mathbf{w}_t, \epsilon_m) = \mathbf{w}_t + \epsilon_m\end{aligned}$$

$$\begin{aligned}\Psi &= \left. \frac{\partial}{\partial \mathbf{w}} \begin{pmatrix} w_1 + \epsilon_{1,p} \\ w_1 \sin(w_1) + \epsilon_{2,p} \end{pmatrix} \right|_{\mu_{t-1}, 0} = \begin{pmatrix} 1 & 0 \\ \sin(\mu_{1,t-1}) + \mu_{1,t-1} \cos(\mu_{1,t-1}) & 0 \end{pmatrix} \\ \Upsilon_p &= \left. \frac{\partial}{\partial \epsilon_p} \begin{pmatrix} w_1 + \epsilon_{1,p} \\ w_1 \sin(w_1) + \epsilon_{2,p} \end{pmatrix} \right|_{\mu_{t-1}, 0} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \Phi &= \left. \frac{\partial}{\partial \mathbf{w}} \begin{pmatrix} w_1 + \epsilon_{1,m} \\ w_2 + \epsilon_{2,m} \end{pmatrix} \right|_{\mu_+, 0} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ \Upsilon_m &= \left. \frac{\partial}{\partial \epsilon_m} \begin{pmatrix} w_1 + \epsilon_{1,m} \\ w_2 + \epsilon_{2,m} \end{pmatrix} \right|_{\mu_+, 0} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}\end{aligned}$$

$$\mathbf{y} = f(\mathbf{x}) \quad \frac{\partial f}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_M} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_N}{\partial x_1} & \cdots & \frac{\partial f_N}{\partial x_M} \end{pmatrix}$$

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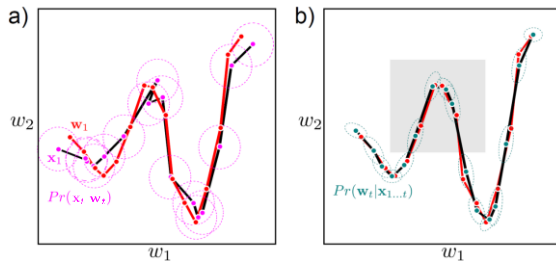
Extended Kalman Filter

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$$\mathbf{w}_t = f(\mathbf{w}_{t-1}, \epsilon_p) = \begin{pmatrix} w_{1,t-1} + \epsilon_{1,p} \\ w_{1,t-1} \sin(w_{1,t-1}) + \epsilon_{2,p} \end{pmatrix}$$

$$\mathbf{x}_t = g(\mathbf{w}_t, \epsilon_m) = \mathbf{w}_t + \epsilon_m$$

Red: True State, Magenta: Observations,
Green: Estimate (Extended Kalman)



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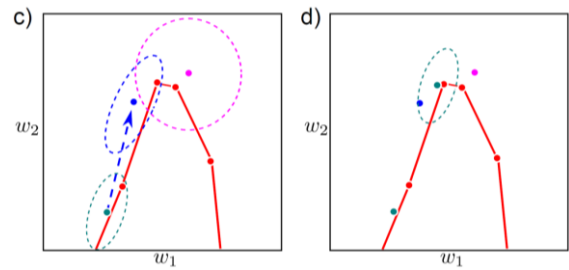
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Extended Kalman Filter

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Red: True State, Magenta: Likelihood, Green: Posterior, Blue: Prediction



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Problems with EKF

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Consider flow field (red) of function f

- Mean is updated by non-linear function without noise

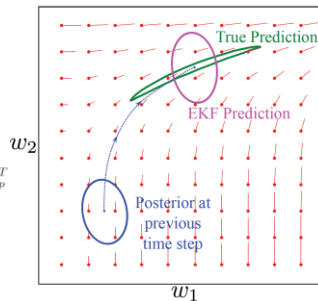
$$\mu_+ = f[\mu_{t-1}, 0]$$

- Covariance is linearly approximated

$$\Sigma_+ = \Psi \Sigma_{t-1} \Psi^T + \Upsilon_p \Sigma_p \Upsilon_p^T$$

- Predicted covariance is poor

How can this be fixed?



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Structure

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- Temporal models
- Kalman filter
- Extended Kalman filter
- Unscented Kalman filter
- Particle filters
- Applications

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Unscented Kalman Filter

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Key ideas:

- Approximate distribution as a sum of weighted particles with correct mean and covariance
- Pass particles through non-linear function of the form

$$\mathbf{w}_t = f[\mathbf{w}_{t-1}] + \epsilon_p$$

$$\mathbf{x}_t = g[\mathbf{w}_t] + \epsilon_m$$
- Compute mean and covariance of transformed variables

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Unscented Kalman Filter

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Approximate with particles where D_w is dimensionality of state:

$$Pr(\mathbf{w}_{t-1} | \mathbf{x}_{1:t-1}) = \text{Norm}_{\mathbf{w}_{t-1}}[\mu_{t-1}, \Sigma_{t-1}]$$

$$\approx \sum_{j=0}^{2D_w} a_j \delta[\mathbf{w}_{t-1} - \hat{\mathbf{w}}^{[j]}]$$

Choose so that

$$\mu_{t-1} = \sum_{j=0}^{2D_w} a_j \hat{\mathbf{w}}^{[j]}$$

$$\Sigma_{t-1} = \sum_{j=0}^{2D_w} a_j (\hat{\mathbf{w}}^{[j]} - \mu_{t-1})(\hat{\mathbf{w}}^{[j]} - \mu_{t-1})^T$$

weights particles or sigma points
 $\sum_{j=0}^{2D_w} a_j = 1$
 $\delta(x) = \begin{cases} 1 & x = 0 \\ 0 & \text{otherwise} \end{cases}$

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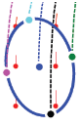
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One possible scheme

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$$\begin{aligned}
 \hat{\mathbf{w}}^{[0]} &= \boldsymbol{\mu}_{t-1} \\
 \hat{\mathbf{w}}^{[j]} &= \boldsymbol{\mu}_{t-1} + \sqrt{\frac{D_{\mathbf{w}}}{1-a_0}} \boldsymbol{\Sigma}_{t-1}^{1/2} \mathbf{e}_j \\
 \hat{\mathbf{w}}^{[D_{\mathbf{w}}+j]} &= \boldsymbol{\mu}_{t-1} - \sqrt{\frac{D_{\mathbf{w}}}{1-a_0}} \boldsymbol{\Sigma}_{t-1}^{1/2} \mathbf{e}_j
 \end{aligned}$$


With:

$$a_j = \frac{1-a_0}{2D_{\mathbf{w}}} \quad \text{Parameter } a_0 \in [0,1]$$

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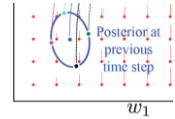
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Parameter

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Parameter a_0

$$\begin{aligned}
 \hat{\mathbf{w}}^{[0]} &= \boldsymbol{\mu}_{t-1} \\
 \hat{\mathbf{w}}^{[j]} &= \boldsymbol{\mu}_{t-1} + \sqrt{\frac{D_{\mathbf{w}}}{1-a_0}} \boldsymbol{\Sigma}_{t-1}^{1/2} \mathbf{e}_j \\
 \hat{\mathbf{w}}^{[D_{\mathbf{w}}+j]} &= \boldsymbol{\mu}_{t-1} - \sqrt{\frac{D_{\mathbf{w}}}{1-a_0}} \boldsymbol{\Sigma}_{t-1}^{1/2} \mathbf{e}_j
 \end{aligned}$$



$a_0 = 0.00$: 68%
 $a_0 = 0.75$: 95.5%

$$\begin{aligned}
 P(\mu - \sigma < X \leq \mu + \sigma) &\approx 68\% \\
 P(\mu - 2\sigma < X \leq \mu + 2\sigma) &\approx 95.5\% \\
 P(\mu - 3\sigma < X \leq \mu + 3\sigma) &\approx 99.7\%
 \end{aligned}$$

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Reconstitution

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Non-linear prediction of sigma points

$$\begin{aligned}
 \hat{\mathbf{w}}_+^{[j]} &= \mathbf{f}[\hat{\mathbf{w}}_+^{[j]}] \\
 \boldsymbol{\mu}_+ &= \sum_{j=0}^{2D_{\mathbf{w}}} a_j \hat{\mathbf{w}}_+^{[j]} \\
 \boldsymbol{\Sigma}_+ &= \sum_{j=0}^{2D_{\mathbf{w}}} a_j (\hat{\mathbf{w}}_+^{[j]} - \boldsymbol{\mu}_+) (\hat{\mathbf{w}}_+^{[j]} - \boldsymbol{\mu}_+)^T + \boldsymbol{\Sigma}_p
 \end{aligned}$$

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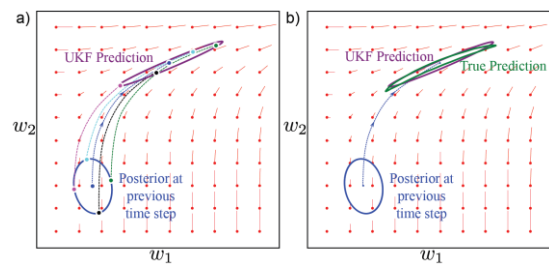
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Unscented Kalman Filter

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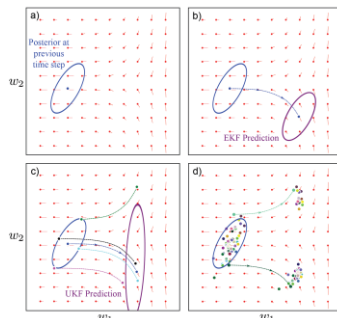
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Problems with UKF

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Structure

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- Temporal models
- Kalman filter
- Extended Kalman filter
- Unscented Kalman filter
- Particle filters
- Applications

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Particle filter



Key idea:

- Represent probability distribution as a set of weighted particles

$$Pr(\mathbf{w}_{t-1} | \mathbf{x}_{1:t-1}) = \sum_{j=1}^J a_j \delta[\mathbf{w}_{t-1} - \hat{\mathbf{w}}_{t-1}^{[j]}]$$

Advantages and disadvantages:

- + Can represent non-Gaussian multimodal densities
- Expensive

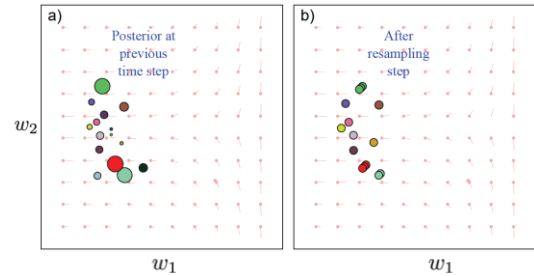
N. Gordon et al. **Novel approach to nonlinear/non-Gaussian Bayesian state estimation**. 1993
M. Isard and A. Blake. **Contour tracking by stochastic propagation of conditional density**. ECCV 1996

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Particle filter



Stage 1: Resample from weighted particles according to their weight to get unweighted particles

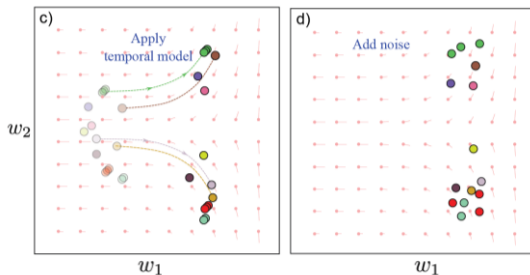


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Particle filter



Stage 2: Pass unweighted samples through temporal model and add noise

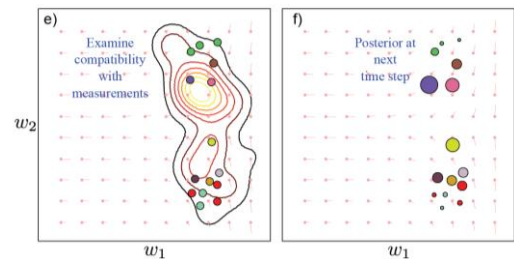


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Particle filter



Stage 3: Weight samples by measurement density



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Generic particle filter



1. Initialisation

- $t \leftarrow 0$
- For $i = 1, \dots, n$, sample $x_{0,0}^{(i)}$ from η_0

2. Prediction

Temporal model

- For $i = 1, \dots, n$, sample $\hat{x}_{t+1,0}^{(i)}$ from $K_t(x_{t,0}^{(i)}, \cdot)$

3. Updating

Likelihood

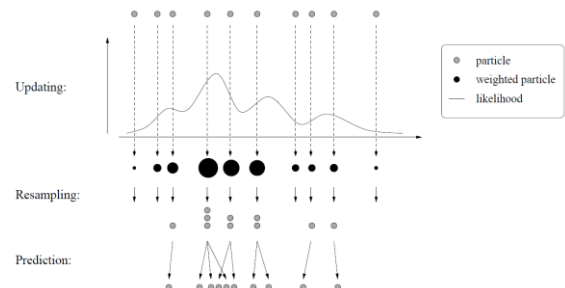
- For $i = 1, \dots, n$, set $\pi_{t+1,0}^{(i)} \leftarrow g_{t+1}(y_{t+1} - h_{t+1}(\hat{x}_{t+1,0}^{(i)}))$
- For $i = 1, \dots, n$, set $\pi_{t+1,0}^{(i)} \leftarrow \frac{\pi_{t+1,0}^{(i)}}{\sum_{j=1}^n \pi_{t+1,0}^{(j)}}$

4. Resampling

- For $i = 1, \dots, n$, set $\hat{x}_{t+1,0}^{(i)} \leftarrow \hat{x}_{t+1,0}^{(j)}$ with probability $\pi_{t+1,0}^{(j)}$
- $t \leftarrow t + 1$ and go to step 2

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Generic particle filter



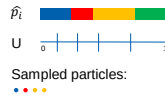
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Resampling



Sample n-times with replacement using probabilities $\sum \pi_i = 1$

- Accumulate weights $\hat{p}_i = \sum_{j \leq i} \pi_j$
- Generate n random numbers [0,1)
- Sort random numbers U
- Collect samples:
 - j=0;
 - for(i=0; i<n; i++)
 - while(U[i] > P[j]) j++;
 - Particle[i] = j;
 - end for



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Convergence



The approximation of the posterior by n particles:

$$\eta_t^n(\omega) := \frac{1}{n} \sum_{i=1}^n \delta_{X_{t,0}^{(i)}(\omega)}$$

Converges (almost surely) to the true posterior as n goes to infinity:

$$P\left(\omega : \eta_t^n(\omega) \xrightarrow{w} \eta_t(\omega)\right) = 1$$

J. Gall. **Generalised Annealed Particle Filter - Mathematical Framework, Algorithms and Applications**. 2005

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Rate of convergence



Convergence rate 1/n for any bounded function ϕ

$$E \left[(\langle \eta_t^n, \phi \rangle - \langle \eta_t, \phi \rangle)^2 \right] \leq c(\epsilon) \frac{\|\phi\|_\infty^2}{n}$$

If temporal model allows to reach any state in a finite number of steps, e.g.,

- True for bounded state space
- Not true for Gaussian noise on infinite state space

Does not depend on dimension of state space, but $c(\epsilon)$ is very large.

In practice, the number of particles is empirical set.

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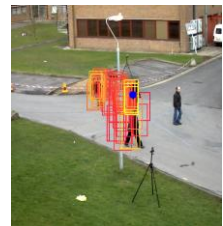
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Tracking pedestrians



- State space 3D (x,y,scale)
- Use Hough transform to compute likelihood for pedestrians
- Initialization by manual marked bounding box



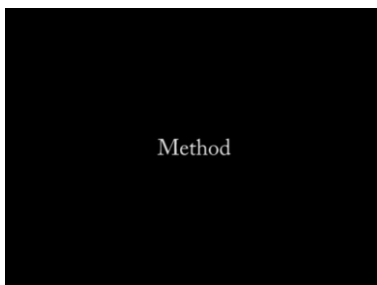
J. Gall et al. **On-line Adaption of Class-specific Codebooks for Instance Tracking**. BMVC 2010
J. Gall et al. **Hough Forests for Object Detection, Tracking, and Action Recognition**. TPAMI 2011

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Results



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Results



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